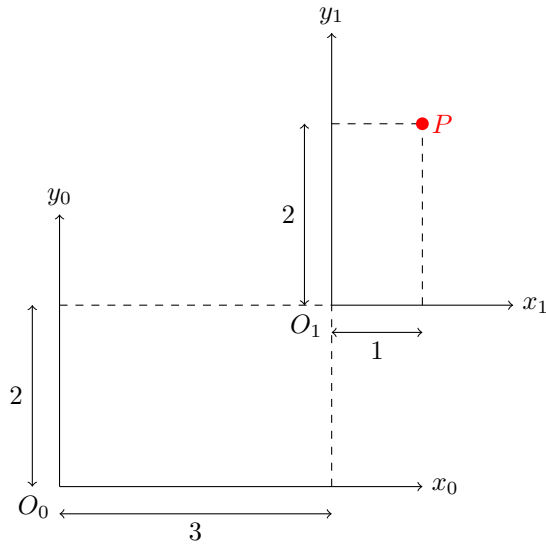


1 July 29, 2025 (Class 1)

2 (Class 2)

3 (Class 3)

3.1 Representing positions



P in frame 0,

$$P^0 = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$$

P in frame 1,

$$P^1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

3.2 Representing rotation in 2D

3.2.1 Representing rotation using θ

3.2.2 Representing rotation using rotation matrix

3.2.3 Representing rotation using direction cosine matrix

For frames $o_0x_0y_0$ and $o_1x_1y_1$, the direction cosine matrix from frame 1 to frame 0 is

$$R_1^0 = \begin{bmatrix} x_1 \cdot x_0 & y_1 \cdot x_0 \\ x_1 \cdot y_0 & y_1 \cdot y_0 \end{bmatrix}$$

Similarly, the direction cosine matrix from frame 0 to frame 1 is

$$R_0^1 = \begin{bmatrix} x_0 \cdot x_1 & y_0 \cdot x_1 \\ x_0 \cdot y_1 & y_0 \cdot y_1 \end{bmatrix}$$

Note that $R_0^1 = (R_1^0)^T$, since it belongs to special orthogonal group $SO(2)$.

3.3 Representing rotation in 3D

3.4 The basic rotation matrices in 3D

$$R_{x,\theta} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta \\ 0 & \sin \theta & \cos \theta \end{bmatrix}$$

$$R_{y,\theta} = \begin{bmatrix} \cos \theta & 0 & \sin \theta \\ 0 & 1 & 0 \\ -\sin \theta & 0 & \cos \theta \end{bmatrix},$$

$$R_{z,\theta} = \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix},$$

3.5 Rotational transformations

Let p^1 be a point in frame 1, and p^0 be the same point in frame 0. Then,

$$p^0 = R_1^0 p^1$$

where R_1^0 is the rotation matrix from frame 1 to frame 0.

3.5.1 Example:

The vector v with coordinates $v^0 = (0, 1, 1)^T$ is rotated about y_0 by $\frac{\pi}{2}$. The resulting vector v_1 has coordinates given by

$$v_1^0 = R_{y,\pi/2} v^0$$

$$= \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ -1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$$

4 (Class 4)

4.1 Similarity transformations

4.2 Composition of rotations

5 (Class 5)

5.1 Euler angles

For a fixed frame $o_0x_0y_0z_0$ and a rotating frame $o_1x_1y_1z_1$, the Euler angles (ϕ, θ, ψ) are defined as follows:

- Rotate frame $o_0x_0y_0z_0$ by ϕ about z_0 to get an intermediate frame $o_ix_iz_i$.

- Rotate frame $o_i x_i y_i z_i$ by θ about y_i to get another intermediate frame $o_j x_j y_j z_j$.
- Rotate frame $o_j x_j y_j z_j$ by ψ about z_j to get the final frame $o_1 x_1 y_1 z_1$.

The resulting rotation matrix from frame 1 to frame 0 is given by

$$\begin{aligned}
 R_1^0 &= R_{z,\phi} R_{y,\theta} R_{z,\psi} = R_{XYZ} \\
 &= \begin{bmatrix} c_\phi & -s_\phi & 0 \\ s_\phi & c_\phi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} c_\theta & 0 & s_\theta \\ 0 & 1 & 0 \\ -s_\theta & 0 & c_\theta \end{bmatrix} \begin{bmatrix} c_\psi & -s_\psi & 0 \\ s_\psi & c_\psi & 0 \\ 0 & 0 & 1 \end{bmatrix} \\
 &= \begin{bmatrix} c_\phi c_\theta c_\psi - s_\phi s_\psi & -c_\phi c_\theta s_\psi - s_\phi c_\psi & c_\phi s_\theta \\ s_\phi c_\theta c_\psi + c_\phi s_\psi & -s_\phi c_\theta s_\psi + c_\phi c_\psi & s_\phi s_\theta \\ -s_\theta c_\psi & s_\theta s_\psi & c_\theta \end{bmatrix}
 \end{aligned}$$

The matrix R_{XYZ} is also known as the ZYZ Euler Angle Transformation. The problem is, given a matrix $R \in SO(3)$,

$$R = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix}$$

determine the Euler angles (ϕ, θ, ψ) , such that $R = R_{XYZ}$.

5.1.1 Solution to the inverse kinematics problem for orientations

5.2 Roll-pitch-yaw angles

5.3 Axis-angle representation

Given a rotation matrix R ,

$$R = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix},$$

we have $(\mathbf{k}, \theta)$, such that

$$\begin{aligned}
 \theta &= \cos^{-1} \left(\frac{r_{11} + r_{22} + r_{33} - 1}{2} \right) \\
 \mathbf{k} &= \frac{1}{2 \sin \theta} \begin{bmatrix} r_{32} - r_{23} \\ r_{13} - r_{31} \\ r_{21} - r_{12} \end{bmatrix}
 \end{aligned}$$

6 August 25, 2025 (Class 6)

6.1 Exponential coordinates

Let M be a $n \times n$ matrix.

We have,

$$e^M = \frac{M^0}{0!} + \frac{M^1}{1!} + \frac{M^2}{2!} + \dots$$

Furthermore,

$$\begin{aligned} e^0 &= I \\ (e^M)^T &= e^{M^T} \\ e^M e^N &= e^{M+N} \quad \text{if } M \text{ and } N \text{ commute} \end{aligned}$$

Let $\mathbf{k} = (k_x, k_y, k_z)^T$ be a 3-dim vector. We define an operator S on $\mathbf{k}$ as,

$$S(\mathbf{k}) = \begin{bmatrix} 0 & -k_z & k_y \\ k_z & 0 & -k_x \\ -k_y & k_x & 0 \end{bmatrix},$$

where $S^T + S = 0$ (S is skew-symmetric matrix).

From this we have,

$$e^{S(\mathbf{k}\theta)} = \mathbf{I} + S(\mathbf{k}\theta) + \frac{S(\mathbf{k}\theta)^2}{2!} + \frac{S(\mathbf{k}\theta)^3}{3!} + \dots$$

6.1.1 Rodrigues' formula

$$e^{S(\mathbf{k})\theta} = \mathbf{I} + \sin(\theta)S(\mathbf{k}) + (1 - \cos(\theta))S^2(\mathbf{k})$$

6.2 Rigid motion

Rigid motion is pure translation together with pure rotation.

6.3 Homogeneous transformations

We have a homogeneous transformation matrix H defined as,

$$H = \begin{bmatrix} R & d \\ 0 & 1 \end{bmatrix}, \quad R \in SO(3), \quad d \in \mathbb{R}^3$$

6.3.1 Elementary Homogeneous transformations

$$\begin{aligned} Trans_{x,a} &= \begin{bmatrix} 1 & 0 & 0 & a \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} & Rot_{x,\alpha} &= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos\alpha & -\sin\alpha & 0 \\ 0 & \sin\alpha & \cos\alpha & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \\ Trans_{y,b} &= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & b \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} & Rot_{y,\beta} &= \begin{bmatrix} \cos\beta & 0 & \sin\beta & 0 \\ 0 & 1 & 0 & 0 \\ -\sin\beta & 0 & \cos\beta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \\ Trans_{z,c} &= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & c \\ 0 & 0 & 0 & 1 \end{bmatrix} & Rot_{z,\gamma} &= \begin{bmatrix} \cos\gamma & -\sin\gamma & 0 & 0 \\ \sin\gamma & \cos\gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \end{aligned}$$

6.3.2 Composition rules for Homogeneous transformations

Given a homogeneous transformation H_1^0 , if a second rigid motion $H \in SE(3)$ is performed,

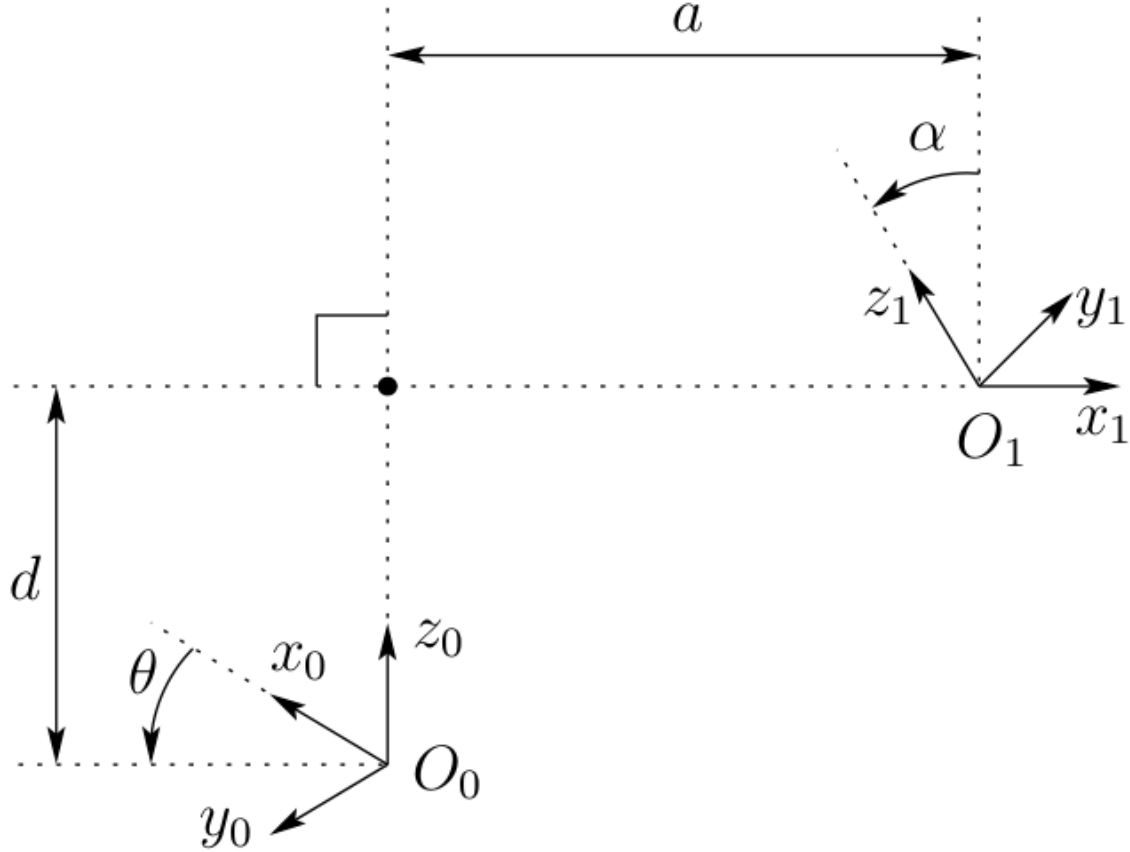
1. relative to the **current** frame, then

$$H_2^0 = H_1^0 H$$

2. relative to the **fixed** frame, then

$$H_2^0 = H H_1^0$$

6.4 Forward kinematics: DH convention



$$A_i = Rot_{z, \theta_i} Trans_{z, d_i} Trans_{x, a_i} Rot_{x, \alpha_i}$$

$$= \begin{bmatrix} c_{\theta_i} & -s_{\theta_i} c_{\alpha_i} & s_{\theta_i} s_{\alpha_i} & a_i c_{\theta_i} \\ s_{\theta_i} & c_{\theta_i} c_{\alpha_i} & -c_{\theta_i} s_{\alpha_i} & a_i s_{\theta_i} \\ 0 & s_{\alpha_i} & c_{\alpha_i} & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$(a_i, \alpha_i, d_i, \theta_i)$ are called **link length**, **link twist**, **link offset**, **joint angle**.

DH Coordinate Frame Assumptions

1. The axis x_1 is perpendicular to the axis z_0 .
2. The axis x_1 intersects the axis z_0 .

6.5 Formulas

6.5.1 Derivative of a rotation matrix

$$\frac{d}{d\theta} R_{k,\theta} = S(k) R_{k,\theta}$$

6.5.2 Angular velocity, general case

$$\dot{R}(t) = S(t) R(t)$$

$$\dot{R}(t) = S(\omega(t)) R(t)$$

6.5.3 Addition of angular velocities

Let

$$R_n^0 = R_1^0 R_2^1 \dots R_n^{n-1}$$

then

$$\dot{R}_n^0 = S(\omega_{0,n}^0) R_n^0$$

where

$$\begin{aligned} \omega_{0,n}^0 &= \omega_{0,1}^0 + R_1^0 \omega_{1,2}^1 + \dots + R_{n-1}^0 \omega_{n-1,n}^{n-1} \\ &= \omega_{0,1}^0 + \omega_{1,2}^0 + \dots + \omega_{n-1,n}^0 \end{aligned}$$

6.5.4 Linear velocity of a point attached to a moving frame

Let f_0 and f_1 be two frames with f_1 moving relative to f_0 , and

$$H_1^0(t) = \begin{bmatrix} R_1^0(t) & o_1^0(t) \\ 0 & 1 \end{bmatrix}$$

be a time dependent homogeneous transformation relating the two frames. Now,

$$p^0 = R_1^0 p^1 + o_1^0$$

and

$$\begin{aligned} \dot{p}^0 &= \dot{R}_1^0 p^1 + \dot{o}_1^0 \\ &= S(\omega) R_1^0 p^1 + \dot{o}_1^0 \\ &= \omega \times r + v \end{aligned}$$

If the point p is moving w.r.t f_1 , Then

$$\dot{p}^0 = \dot{R}_1^0 p^1 + R_1^0 \dot{p}^1 + \dot{o}_1^0$$

4.6.3 Combining the Linear and Angular Velocity Jacobians

As we have seen in the preceding section, the upper half of the Jacobian J_v is given as

$$J_v = [J_{v_1} \cdots J_{v_n}] \quad (4.55)$$

in which the i^{th} column J_{v_i} is

$$J_{v_i} = \begin{cases} z_{i-1} \times (o_n - o_{i-1}) & \text{for revolute joint } i \\ z_{i-1} & \text{for prismatic joint } i \end{cases} \quad (4.56)$$

The lower half of the Jacobian is given as

$$J_\omega = [J_{\omega_1} \cdots J_{\omega_n}] \quad (4.57)$$

in which the i^{th} column J_{ω_i} is

$$J_{\omega_i} = \begin{cases} z_{i-1} & \text{for revolute joint } i \\ 0 & \text{for prismatic joint } i \end{cases} \quad (4.58)$$

The above formulas make the determination of the Jacobian of any manipulator simple since all of the quantities needed are available once the forward kinematics are worked out. Indeed, the only quantities needed to compute the Jacobian are the unit vectors z_i and the coordinates of the origins $o_1, \dots, o_n$. A moment's reflection shows that the coordinates for z_i with respect to the base frame are given by the first three elements in the third column of T_i^0 while o_i is given by the first three elements of the fourth column of T_i^0 . Thus, only the third and fourth columns of the T matrices are needed in order to evaluate the Jacobian according to the above formulas.

6.5.5 See also

6.5.6 References